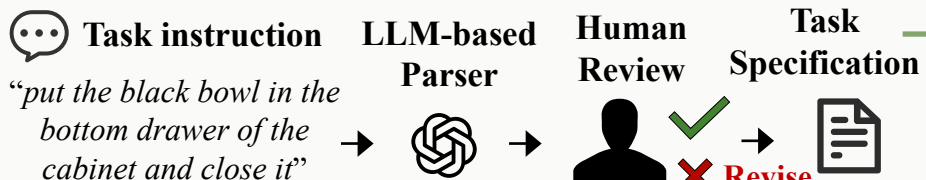
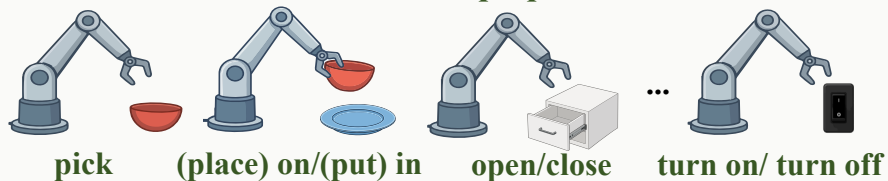


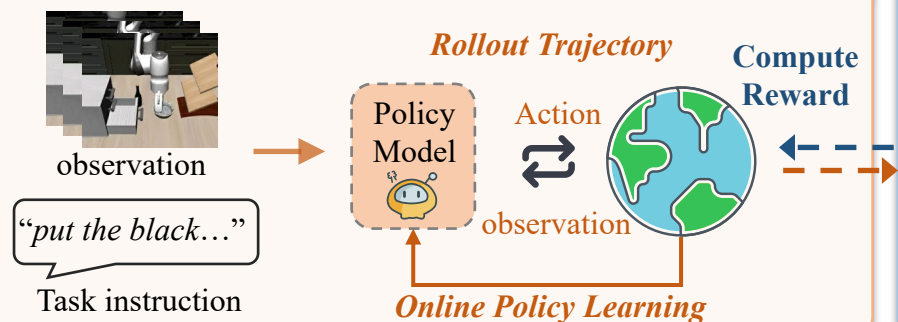
(1) Natural-Language Instruction--STL Specification (Offline)

Task-level atomic proposition

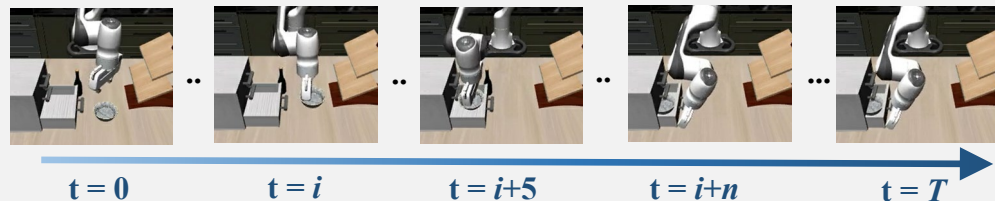


$$\varphi = \Diamond_{[0,T]} \left(\text{pick}(\text{black_bowl}) \wedge \Diamond_{[0,T]} \left(\text{in}(\text{black_bowl}, \text{bottom_drawer}) \wedge \Diamond_{[0,T]} \text{close}(\text{bottom_drawer}) \right) \right).$$

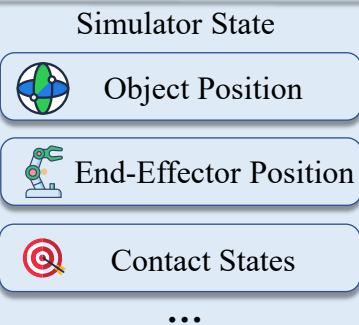
(3) RLFT with STL-Based Rewards (Online)



(2) STL Specification Robustness Monitoring (Online)



Privileged State Estimator



Vision-Based Estimator

